JOHOR MATRICULATION ENGINEERING COLLEGE

MATHLETICS SEM 1

SESI 2023/2024

MATHEMATICS 1

INSTRUCTIONS TO STUDENTS

1. *There are TWO sections in the question paper. Section A and Section B.*
2. ***Section A** consists of **3** questions. **Section B** consists of **7** questions.*
3. *Answer all questions.*

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Trigonometry
Trigonometri

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin\frac{A+B}{2} \cos\frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos\frac{A+B}{2} \sin\frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos\frac{A+B}{2} \cos\frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin\frac{A+B}{2} \sin\frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos^2 A - \sin^2 A \\ &= 2\cos^2 A - 1 \\ &= 1 - 2\sin^2 A \end{aligned}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

LIST OF MATHEMATICAL FORMULAE
SENARAI RUMUS MATEMATIK

Quadratic equation $ax^2+bx+c=0$:
Persamaan kuadratik $ax^2+bx+c=0$:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Differentiation
Pembezaan

$f(x)$	$f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

$$\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$$

$$\frac{d^2y}{dx^2}=\frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Sphere ***Sfera***	$V=\frac{4}{3}\pi r^3$	$S=4\pi r^2$
Right circular cone ***Kon membulat tegak***	$V=\frac{1}{3}\pi r^2 h$	$S=\pi r^2+\pi rl$
Right circular cylinder ***Silinder membulat tegak***	$V=\pi r^2 h$	$S=2\pi r^2+2\pi rh$

SECTION A [25 *marks*]

BAHAGIAN **A** [25 *markah*]

This section consists of **3** questions. Answer **all** questions.

Bahagian ini mengandungi **3** *soalan. Jawab* ***semua*** *soalan.*

1 Evaluate the following limits, if they exist.

(a) $\lim\limits_{x \to 3} \sqrt[3]{\dfrac{x^2 - 1}{2x^3 + x^2 + 1}}$ [3 *marks*]

(b) Find the value of m if $\lim\limits_{x \to 0} \dfrac{3x^2 + mx}{4x - 4x^2} = 2.$ [3 *marks*]

(c) $\lim\limits_{x \to -\infty} \sqrt{\dfrac{x^2 - 1}{2x^2 + 2x + 1}}$ [3 *marks*]

2 Find $\dfrac{dy}{dx}$ for each of the following

(a) $y = x^{\frac{1}{2}} 2^{\sin x}$ [4 *marks*]

(b) $y = \ln\left(x\sqrt{1 + x^2}\right)$ [5 *marks*]

3 Show that the function $f(x) = 2x - 6x^{\frac{1}{3}}$ has three critical points. State the coordinates of each of the critical points.

[7 *marks*]

SECTION B [75 *marks*]

***BAHAGIAN* B** [75 *markah*]

This section consists of **7** questions. Answer **all** questions.

Bahagian ini mengandungi **7** *soalan. Jawab* ***semua*** *soalan.*

1 Given $z_1 = 3+4i$ and $z_2 = 5-2i$. If $\frac{1}{z_3} = \frac{1}{z_1} + \frac{1}{z_1 z_2}$ find z_3 in the form of $a+bi$.

[5 *marks*]

2 (a) Solve $x\ln 3 = 2 - x\ln 2$.

[*4 marks*]

(b) Determine the solution set of the inequality $\left|\frac{2x-1}{x+3}\right| \leq 1$. [*7 marks*]

3 A piecewise function is defined as $f(x) = \begin{cases} \frac{4x-9}{3} & , \quad x \leq 3 \\ 1+\sqrt{x-3} & , \quad 3 < x < 7 \\ 3+2(x-7)^2 & , \quad x \geq 7 \end{cases}$

(a) Find *f*(5). [1 *mark*]

(b) Sketch *f(x)* and $f^{-1}(x)$ on the same axes. Hence, state the domain and range of *f(x)* and $f^{-1}(x)$. [5 *marks*]

4 (a) Given $f(x) = \frac{e^{2x}+1}{e^{2x}-1}$ and $g(x) = \frac{2e^x}{e^{2x}-1}$.

(i) Find the value of *a* such that $[f(x)]^2 - [g(x)]^2 = a$. [4 *marks*]

(ii) Show that inverse of *f* is $\frac{1}{2}\ln\left|\frac{1+x}{x-1}\right|$. [3 *marks*]

(b) Given $h(x)=-\left(\frac{1}{3}\right)^{2x-3}-1$.

(i) Find $h^{-1}(x)$. [4 *marks*]

(ii) Sketch $h(x)$ and $h^{-1}(x)$ on the same axes. [3 *marks*]

5 A function *f* is defined by

$$f(x)=\begin{cases}\dfrac{2+x}{4-x^2}, & x<2\\ k, & 2\le x<3\\ x-2k, & x\ge 3\end{cases}.$$

(a) State the definition of the continuity of a function at a point. [2 *marks*]

(b) Determine the value of *k* if the function $f(x)$ is continuous at $x=3$.

[2 *marks*]

(c) Find the vertical asymptote. [3 *marks*]

(d) Find the horizontal asymptote. [4 *marks*]

(e) Sketch the graph of $f(x)$. [3 *marks*]

6 (a) If $\boldsymbol{y}=\boldsymbol{e}^{-4x}\,\mathbf{ln}(2\boldsymbol{x})$, show that $\boldsymbol{e}^{4x}\left(\frac{\boldsymbol{d}^2\boldsymbol{y}}{\boldsymbol{dx}^2}+4\frac{\boldsymbol{dy}}{\boldsymbol{dx}}\right)=\frac{-(1+4\boldsymbol{x})}{\boldsymbol{x}^2}$. [9 *marks*]

(b) Given that $\boldsymbol{x}=\frac{1}{\boldsymbol{t}^2-1}$ and $\boldsymbol{y}=\frac{1+\boldsymbol{t}^2}{2\boldsymbol{t}}$, where $\boldsymbol{t}>0$.

Find the derivative of *y* upon *x*. Hence find the value of *t* when the derivative is zero.

[7 *marks*]

7 P and Q are two resistors. The total resistance, R, is given by the formula:

$$\frac{1}{R}=\frac{1}{P}+\frac{1}{Q}$$

The resistances of P and Q are increasing at the rates of 0.01-ohm s^{-1} and 0.03-ohm s^{-1} respectively.

(a) Write an equation involving the related rates, $\frac{dP}{dt}$, $\frac{dQ}{dt}$ and $\frac{dR}{dt}$.

[3 *marks*]

(b) Calculate the rate at which R is changing at the instant when $P=30$ ohms and $Q=60$ ohms.

[6 *marks*]

END OF QUESTION PAPER

KERTAS SOALAN TAMAT